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7. SURFACE TENSION BY CAPILLARY RISE METHOD

AIM

To determine surface tension of a liquid by capillary rise method.

APPARATUS REQUIRED

A beaker of Water, capillary tube, vernier microscope, two holed rubber stopper, a knitting needle, a short rubber tubing and retort clamp.

FORMULA

The surface tension of the liquid $T = \frac{hr\sigma g}{2}$ $N\,m^{-1}$

where $T \rightarrow$ Surface tension of the liquid $(N\,m^{-1})$

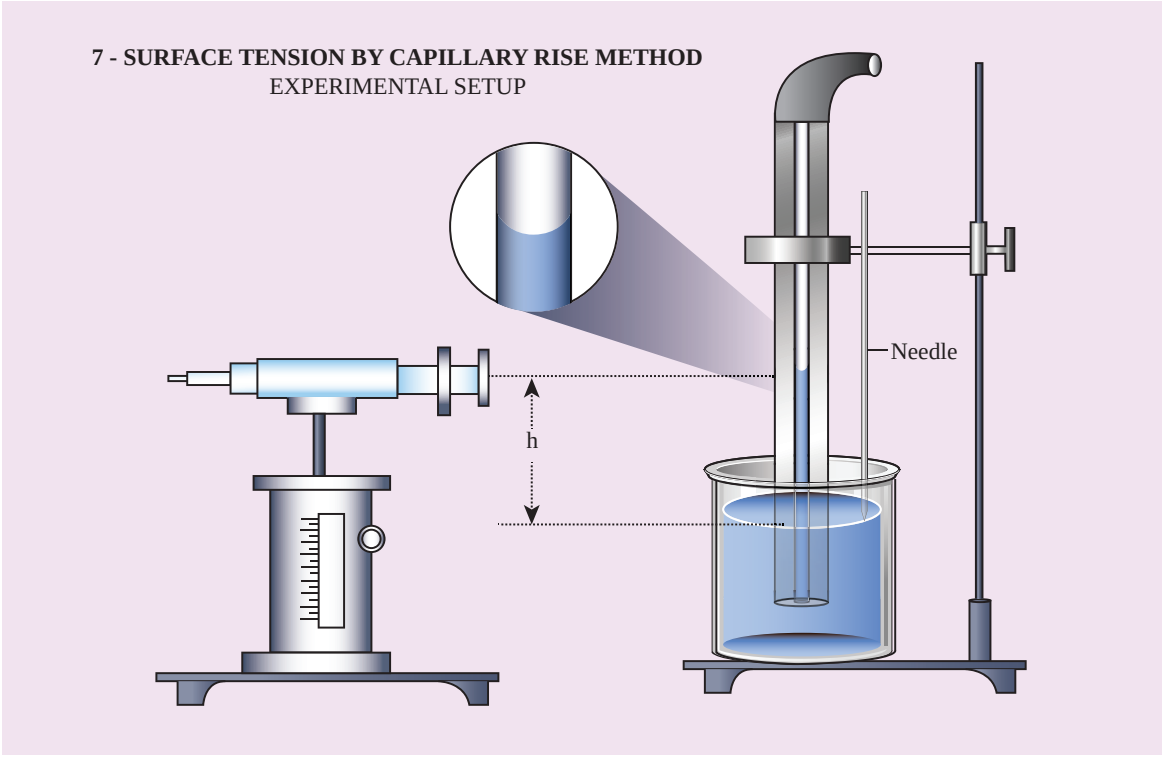
$h \rightarrow$ height of the liquid in the capillary tube (m)

$r \rightarrow$ radius of the capillary tube (m)

$\sigma \rightarrow$ Density of water $(kg\,m^{-3})$ $(\sigma = 1000\ kg\,m^{-3})$

$g \rightarrow$ Acceleration due to gravity (g = 9.8 m s^{-2})

DIAGRAM



PROCEDURE

- A clean and dry capillary tube is taken and fixed in a stand
- A beaker containing water is placed on an adjustable platform and the capillary tube is dipped inside the beaker so that a little amount of water is raised inside.